

WASP-104b

R-2016

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_260415 · created 2026-08-07T23:42:08+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2461145.6968 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.024
Transit depth (Rp/Rs)^2	2.0 +/- 0.69 [%]
Orbital Inclination (inc)	84.44 +/- 1.34
Airmass coefficient 1 (a1)	1.209 +/- 0.019
Airmass coefficient 2 (a2)	-0.156 +/- 0.05
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	None
Optimal Aperture	4.86
Optimal Annulus	7.7
Transit Duration (day)	0.0691 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.024 vs 0.1214 (deviation 0.5σ)
Duration fit vs published	0.0691 d vs 0.0742 d (deviation 0.49σ)
Mid-time O–C	-15.3 min
Depth SNR	3.4
Residual scatter	1.53 %

Light curve

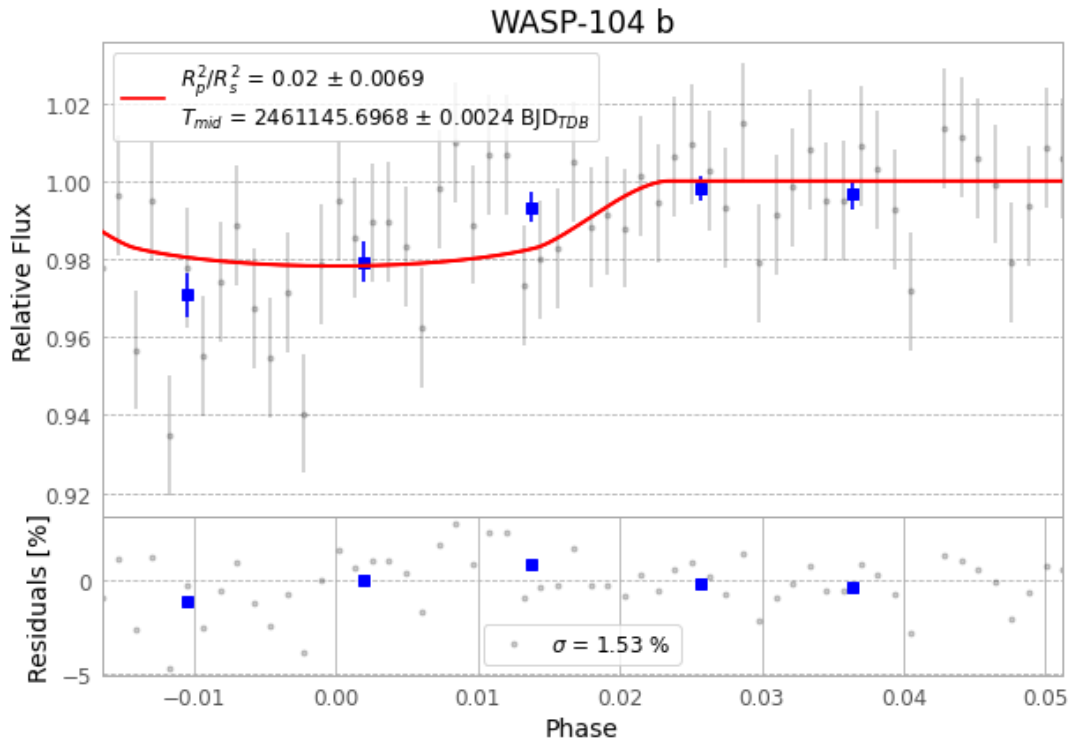


Figure 1 — FinalLightCurve_WASP-104 b_2026-04-14.png (SHA-256 66d0248499a92733...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [303, 166], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c755f97cbe63ad56...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

58 FITS frames (38 MB), WASP-104260415035416.FITS → WASP-104260415064516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-260415.log (34383af51b43...), FinalLightCurve_WASP-104 b_2026-04-14.png (66d0248499a9...), FinalLightCurve_WASP-104 b_2026-04-14.csv (2d0c6b85cf51...)

Compute resources

Wall time 303.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 23:42Z.